Can the $n = 2$ response function correct interference?

1 Setup and question

A model has response $R(v) = F(S(v))$ with

$$S(v) = \sum_{i=1}^{M} w_i (d_i^\top v)^2 = v^\top A v, \qquad A = \sum_i w_i d_i d_i^\top,$$

unit feature directions $\{d_i\}$ in $\mathbb{R}^D$ with $M > D$, weights $w_i \geq 0$, and plateau $\{v : S(v) < c\}$. The plateau-breaking magnitude along unit d is $m_d = \sqrt{c/(d^\top A d)}$. Write $q_i := d_i^\top A d_i$.

With k active features, an inactive feature i picks up spurious read-off of typical size $\sigma_a\sqrt{k/D}$. We demand:

- **Per-feature correction:** $\sigma_a\sqrt{k/D} \leq m_{d_i} \leq \sigma_a$ for every i — i.e. $c/\sigma_a^2 \leq q_i \leq Dc/(k\sigma_a^2)$.
- **LRH sensitivity:** $m_{d_i} < m_{d_r}$ for every feature i (random unit d_r) — i.e. $q_i > \mathrm{tr}(A)/D$.

Question. Do weights $\{w_i\}$ and $c > 0$ exist making both conditions hold simultaneously?

2 Answer (i): under tight framing, no

If the feature frame is tight ($\Phi\Phi^\top = (M/D)\, I$), the answer is **no, for any weight choice**.

Summing LRH over features:

$$\sum_i q_i = \mathrm{tr}(A\, \Phi\Phi^\top) = \frac{M}{D}\ \mathrm{tr}\, A.$$

So the average of q_i over features equals $\mathrm{tr}(A)/D$ exactly — at least one feature must violate strict LRH, regardless of $\{w_i\}$. No reweighting rescues a tight frame.

Why tight framing is the usual assumption. Tight frames minimise worst-case coherence $\max_{i\neq j} |G_{ij}|$ (Welch bound), equalise the spurious read-offs $\hat{a}_i$ across features, and make $\Phi\Phi^\top$ isotropic so that random directions in activation space are all treated the same. These are exactly the conditions under which the typical-interference estimate $\sigma_a\sqrt{k/D}$ is valid. Tightness is therefore the natural idealisation whenever one wants interference to be uniformly small across features — and the $n = 2$ response function cannot simultaneously satisfy interference correction and LRH sensitivity in this idealisation.

3 Answer (ii): lifting tight framing, yes

If the frame is not tight, the answer is **yes**. The simplest construction: take directions with $G_{ij}^2 \approx 1/D$ for $i \neq j$ (i.e. the regime in which the $\sigma_a\sqrt{k/D}$ estimate is accurate, but without enforcing $\Phi\Phi^\top = (M/D)I$). With uniform weights $w_i = 1$,

$$q_i = 1 + \sum_{j \neq i} G_{ij}^2 \;\approx\; \frac{M}{D} + \frac{D-1}{D}.$$

Then $q_i - \mathrm{tr}(A)/D \approx (D-1)/D$: a margin of order 1 per feature, so LRH holds for every i. Choose $c = \sigma_a^2 \min_i q_i$: the feature with smallest q_i hits $m = \sigma_a$; all others satisfy $q_i \in [q_{\min}, q_{\max}]$ with $q_{\max}/q_{\min} = 1 + O(\sqrt{M}/D)$, easily inside the factor-D/k bracket for $k \ll D$.

What this costs. Dropping tight framing buys feasibility but introduces a parasitic privileged direction. The top eigenvector u of $\Phi\Phi^\top$ has $q_u = \lambda_{\max}(A) > q_i$ for every feature, so u — which is *not* a feature direction — is steeper than any feature. LRH then holds only in the typical sense (features beat typical random directions), not universally (features do not beat the worst-case random direction). Non-uniform interference also appears: feature pairs with G_{ij}^2 above average exceed the Welch bound, and the $\sigma_a\sqrt{k/D}$ estimate becomes an approximation rather than a uniform bound.

4 Summary

- Tight frame $\Rightarrow$ LRH and per-feature correction cannot coexist, for any weights.
- Non-tight frame with $G_{ij}^2 \approx 1/D$ uniformly $\Rightarrow$ both conditions hold with uniform weights.
- The cost of dropping tightness is a non-feature direction (the top eigenvector of $\Phi\Phi^\top$) that is strictly steeper than any feature.